

CHAPTER 4 – EXPLORING ALGEBRAIC IDENTITIES

EXERCISE SET 4.5

IMPORTANT IDENTITIES

$$a^3 - b^3 = (a - b)(a^2 + ab + b^2)$$

$$a^3 + b^3 = (a + b)(a^2 - ab + b^2)$$

$$a^3 + b^3 + c^3 - 3abc = (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca)$$

$$a^2 - b^2 = (a - b)(a + b)$$

$$(a - b)^2 = a^2 - 2ab + b^2$$

1. Simplify the following rational expressions, assuming that the expressions in the denominators are not equal to zero.

(i) $\frac{3p^2 - 3pq - 18q^2}{p^2 + 3pq - 10q^2}$

(ii) $\frac{n^3 - 3n^2m + 3nm^2 - m^3}{5m^2 - 10mn + 5n^2}$

(iii) $\frac{w^3 - v^3 + x^3 + 3wvx}{w^2 + v^2 + x^2 - 2wv - 2vx + 2wx}$

(iv) $\frac{4y^2 - 20yz + 25z^2}{25z^2 - 4y^2}$

(v) $\frac{(x^2 + x - 6)(x^2 - 7x + 12)}{(x^2 - 6x + 8)(x^2 - 9)}$

(vi) $\frac{p^4 - 16}{p^2 - 4p + 4}$

(i)

$$\frac{3p^2 - 3pq - 18q^2}{p^2 + 3pq - 10q^2}$$

Factorise the numerator:

$$\begin{aligned} 3p^2 - 3pq - 18q^2 &= 3(p^2 - pq - 6q^2) \\ &= 3(p - 3q)(p + 2q). \end{aligned}$$

Factorise the denominator:

$$\begin{aligned} p^2 + 3pq - 10q^2 \\ &= p^2 + 5pq - 2pq - 10q^2 \\ &= p(p + 5q) - 2q(p + 5q) \\ &= (p + 5q)(p - 2q). \end{aligned}$$

Therefore,

$$\boxed{\frac{3(p - 3q)(p + 2q)}{(p + 5q)(p - 2q)}}$$

No common factor can be cancelled.

(ii)

$$\frac{n^3 - 3n^2m + 3nm^2 - m^3}{5m^2 - 10mn + 5n^2}$$

Using

$$(a - b)^3 = a^3 - 3a^2b + 3ab^2 - b^3,$$

we get

$$n^3 - 3n^2m + 3nm^2 - m^3 = (n - m)^3.$$

Now,

$$\begin{aligned} 5m^2 - 10mn + 5n^2 &= 5(m^2 - 2mn + n^2) \\ &= 5(m - n)^2 = 5(n - m)^2. \end{aligned}$$

Hence,

$$\frac{(n - m)^3}{5(n - m)^2} = \boxed{\frac{n - m}{5}}.$$

(iii)

$$\frac{w^3 - v^3 + x^3 + 3wvx}{w^2 + v^2 + x^2 - 2wv - 2vx + 2wx}$$

Take

$$a = w, \quad b = -v, \quad c = x.$$

Using

$$a^3 + b^3 + c^3 - 3abc = (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca),$$

we get

$$\begin{aligned} & w^3 - v^3 + x^3 + 3wvx \\ &= (w - v + x)(w^2 + v^2 + x^2 + wv + vx - wx). \end{aligned}$$

Also,

$$\begin{aligned} & w^2 + v^2 + x^2 - 2wv - 2vx + 2wx \\ &= (w - v + x)^2. \end{aligned}$$

Therefore,

$$\begin{aligned} & \frac{(w - v + x)(w^2 + v^2 + x^2 + wv + vx - wx)}{(w - v + x)^2} \\ &= \boxed{\frac{w^2 + v^2 + x^2 + wv + vx - wx}{w - v + x}}. \end{aligned}$$

(iv)

$$\frac{4y^2 - 20yz + 25z^2}{25z^2 - 4y^2}$$

Factorise the numerator:

$$4y^2 - 20yz + 25z^2 = (2y - 5z)^2.$$

Factorise the denominator:

$$\begin{aligned} 25z^2 - 4y^2 &= (5z)^2 - (2y)^2 \\ &= (5z - 2y)(5z + 2y). \end{aligned}$$

Since

$$5z - 2y = -(2y - 5z),$$

we get

$$\begin{aligned} &\frac{(2y - 5z)^2}{-(2y - 5z)(5z + 2y)} \\ &= -\frac{2y - 5z}{5z + 2y}. \end{aligned}$$

Therefore,

$$\boxed{\frac{5z - 2y}{5z + 2y}}$$

(v)

$$\frac{(x^2 + x - 6)(x^2 - 7x + 12)}{(x^2 - 6x + 8)(x^2 - 9)}$$

Factorise each quadratic:

$$x^2 + x - 6 = (x + 3)(x - 2),$$

$$x^2 - 7x + 12 = (x - 3)(x - 4),$$

$$x^2 - 6x + 8 = (x - 2)(x - 4),$$

and

$$x^2 - 9 = (x - 3)(x + 3).$$

Therefore,

$$\frac{(x + 3)(x - 2)(x - 3)(x - 4)}{(x - 2)(x - 4)(x - 3)(x + 3)}.$$

Cancelling the common factors,

$$\boxed{1}.$$

(vi)

$$\frac{p^4 - 16}{p^2 - 4p + 4}$$

Factorise the numerator using difference of squares:

$$\begin{aligned} p^4 - 16 &= (p^2)^2 - 4^2 \\ &= (p^2 - 4)(p^2 + 4). \end{aligned}$$

Again,

$$p^2 - 4 = (p - 2)(p + 2).$$

Therefore,

$$p^4 - 16 = (p - 2)(p + 2)(p^2 + 4).$$

Now factorise the denominator:

$$p^2 - 4p + 4 = (p - 2)^2.$$

Hence,

$$\begin{aligned} &\frac{(p - 2)(p + 2)(p^2 + 4)}{(p - 2)^2} \\ &= \boxed{\frac{(p + 2)(p^2 + 4)}{p - 2}}. \end{aligned}$$

EXERCISE 4.5 – FINAL ANSWERS

$$(i) \quad \frac{3(p-3q)(p+2q)}{(p+5q)(p-2q)}$$

$$(ii) \quad \frac{n-m}{5}$$

$$(iii) \quad \frac{w^2 + v^2 + x^2 + wv + vx - wx}{w - v + x}$$

$$(iv) \quad \frac{5z-2y}{5z+2y}$$

$$(v) \quad 1$$

$$(vi) \quad \frac{(p+2)(p^2+4)}{p-2}$$

QUICK REVISION

$$a^2 - b^2 = (a-b)(a+b)$$

$$a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2)$$

$$a^3 + b^3 + c^3 - 3abc = (a+b+c)(a^2 + b^2 + c^2 - ab - bc - ca)$$

Simplification rule:

First factorise the numerator and denominator, then cancel only the common factors.